

Computer Science 2

SG 7 - Validating Data Input and Verifying Output

Hands-on Activity

Instructions: Write a Python program for each problem and apply the appropriate input validation techniques. Test your programs using the sample input and expected output as guides.

Create one Python source file for each problem using the filenames **prob1.py**, **prob2.py**, and **prob3.py**.

Record one video explanation for each problem. In each video, explain the validation techniques used and how your code works. Use the filenames **prob1video**, **prob2video**, and **prob3video**.

Upload all three Python source files and three video explanations to your **GitHub repository**. Submit only the link to your **GitHub repository in KHub** on or before **11:59 PM, September 22, 2026**.

1. Student Age Validator

Create a Python program that asks the user to enter their **age**. Check whether the entered age is a whole number and accept only ages from **12 to 18**, inclusive. Display **Valid age**. if both conditions are satisfied. Display **Invalid age. Age must be from 12 to 18**. if the number is outside the allowed range. Display **Invalid input. Please enter a whole number**. if the input cannot be converted to an integer.

Sample	Sample Input	Expected Output
1	Enter your age: 15	Valid age.
2	Enter your age: fifteen	Invalid input. Please enter a whole number.

2. Username Validator

Create a Python program that asks the user to create a **username**. Accept a username containing **5 to 10 characters**, inclusive. Allow letters and numbers only. Do not allow spaces or special characters. Display **Valid username**. if all requirements are satisfied; otherwise, display **Invalid username**.

Sample	Sample Input	Expected Output
1	Enter username: Juan123	Valid username.
2	Enter username: Juan@123	Invalid username.

3. School Grade Level Validator

Create a Python program that asks the user to enter their **grade level**. Check whether the input is a whole number. Accept only **7, 8, 9, 10, 11, or 12**. Display **Valid grade level**. if the input is accepted, **Invalid grade level**. if the number is not one of the accepted values, and **Invalid input. Please enter a whole number**. if the input cannot be converted to an integer.

Sample	Sample Input	Expected Output
1	Enter your grade level: 8	Valid grade level.
2	Enter your grade level: 6	Invalid grade level.